

Hyperfine structure and isotope shift

6

6.1 Hyperfine structure

Up to this point we have regarded the nucleus as an object of charge $+Ze$ and mass M_N , but it has a magnetic moment μ_I that is related to the nuclear spin $\mathbf{I}$ by

$$\mu_I = g_I \mu_N \mathbf{I}. \quad (6.1)$$

Comparing this to the electron's magnetic moment $-g_s \mu_B \mathbf{s}$ we see that there is no minus sign.¹ Nuclei have much smaller magnetic moments than electrons; the nuclear magneton μ_N is related to the Bohr magneton μ_B by the electron-to-proton mass ratio:

$$\mu_N = \mu_B \frac{m_e}{M_p} \simeq \frac{\mu_B}{1836}. \quad (6.2)$$

The interaction of μ_I with the magnetic flux density created by the atomic electrons $\mathbf{B}_e$ gives the Hamiltonian

$$H_{\text{HFS}} = -\mu_I \cdot \mathbf{B}_e. \quad (6.3)$$

This gives rise to **hyperfine structure** which, as its name suggests, is smaller than fine structure. Nevertheless, it is readily observable for isotopes that have a nuclear spin ($I \neq 0$).

The magnetic field at the nucleus is largest for s-electrons and we shall calculate this case first. For completeness the hyperfine structure for electrons with $l \neq 0$ is also briefly discussed, as well as other effects that can have similar magnitude.

6.1.1 Hyperfine structure for s-electrons

We have previously considered the atomic electrons as having a charge distribution of density $-e|\psi(r)|^2$, e.g. in the interpretation of the direct integral in helium in eqn 3.15 (see also eqn 6.22). To calculate magnetic interactions we need to consider an s-electron as a distribution of magnetisation given by

$$\mathbf{M} = -g_s \mu_B \mathbf{s} |\psi(r)|^2. \quad (6.4)$$

This corresponds to the total magnetic moment of the electron $-g_s \mu_B \mathbf{s}$ spread out so that each volume element $d^3\mathbf{r}$ has a fraction $|\psi(r)|^2 d^3\mathbf{r}$

6.1 Hyperfine structure	97
6.2 Isotope shift	105
6.3 Zeeman effect and hyperfine structure	108
6.4 Measurement of hyperfine structure	112
Further reading	119
Exercises	120

¹Nuclear magnetic moments can be parallel, or anti-parallel, to $\mathbf{I}$, i.e. g_I can have either sign depending on the way that the spin and orbital angular momenta of the protons and neutrons couple together inside the nucleus. The proton (that forms the nucleus of a hydrogen atom) has $g_p > 0$ because of its positive charge. More generally, nuclear magnetic moments can be predicted from the shell model of the nucleus.

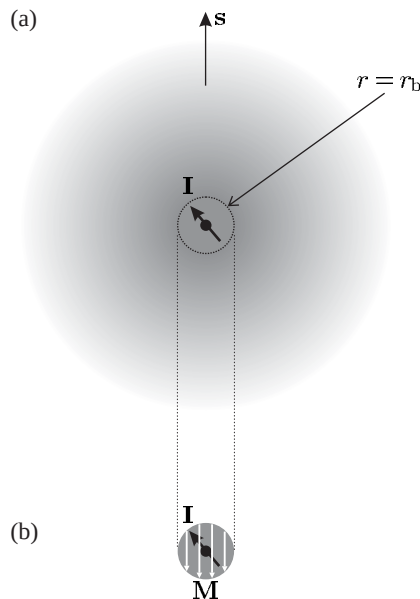


Fig. 6.1 (a) An illustration of a nucleus with spin $\mathbf{I}$ surrounded by the spherically-symmetric probability distribution of an s-electron. (b) That part of the s-electron distribution in the region $r < r_b$ corresponds to a sphere of magnetisation $\mathbf{M}$ anti-parallel to the spin $\mathbf{s}$.

²See Blundell (2001) or electromagnetism texts.

of the total. For s-electrons this distribution is spherically symmetric and surrounds the nucleus, as illustrated in Fig. 6.1. To calculate the field at $r = 0$ we shall use the result from classical electromagnetism² that inside a *uniformly* magnetised sphere the magnetic flux density is

$$\mathbf{B}_e = \frac{2}{3}\mu_0 \mathbf{M}. \quad (6.5)$$

However, we must be careful when applying this result since the distribution in eqn 6.4 is not uniform—it is a function of r . We consider the spherical distribution in two parts.

- (a) A sphere of radius $r = r_b$, where $r_b \ll a_0$ so that the electronic wavefunction squared has a constant value of $|\psi(0)|^2$ throughout this inner region, as indicated on Fig. 6.2.³ From eqn 6.5 the field inside this uniformly magnetised sphere is

$$\mathbf{B}_e = -\frac{2}{3}\mu_0 g_s \mu_B |\psi_{ns}(0)|^2 \mathbf{s}. \quad (6.6)$$

- (b) The part of the distribution outside the sphere $r > r_b$ produces no field at $r = 0$, as shown by the following argument. Equation 6.5 for the field inside a sphere does not depend on the radius of that sphere—it gives the same field for a sphere of radius r and a sphere of radius $r + dr$. Therefore the contribution from each shell of thickness dr is zero. The region $r > r_b$ can be considered as being made up of many such shells that give no additional contribution to the field.⁴

Putting this field and μ_I from eqn 6.1 into eqn 6.3 gives

$$H_{\text{HFS}} = g_I \mu_N \mathbf{I} \cdot \frac{2}{3}\mu_0 g_s \mu_B |\psi_{ns}(0)|^2 \mathbf{s} = A \mathbf{I} \cdot \mathbf{s}. \quad (6.7)$$

³This spherical boundary at $r = r_b$ does not correspond to anything physical in the atom but is chosen for mathematical convenience. The radius r_b should be greater than the radius of the nucleus r_N and it easy to fulfil the conditions $r_N \ll r_b \ll a_0$ since typical nuclei have a size of a few fermi (10^{-15} m), which is five orders of magnitude less than atomic radii.

⁴The magnetisation is a function of r only and therefore each shell has a uniform magnetisation $\mathbf{M}(r)$ between r and $r + dr$. The proof that these shells do not produce a magnetic flux density at $r = 0$ does not require $\mathbf{M}(r)$ to be the same for all the shells, and clearly this is not the case. Alternatively, this result can be obtained by integrating the contributions to the field at the origin from the magnetic moments $\mathbf{M}(r) d^3\mathbf{r}$ over all angles (θ and ϕ).

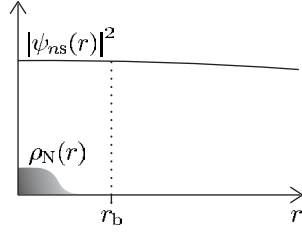


Fig. 6.2 (a) The probability density $|\psi(r)|^2$ of an s-electron at small distances ($r \ll a_0$) is almost constant. The distribution of nuclear matter $\rho_N(r)$ gives an indication of the nuclear radius r_N . To calculate the interaction of the nuclear magnetic moment with an s-electron the region is divided into two parts by a boundary surface of radius $r = r_b \gg r_N$ (as also shown in Fig. 6.1). The inner region corresponds to a sphere of uniform magnetisation that produces a flux density $\mathbf{B}_e$ at $r = 0$. The nuclear magnetic moment interacts with this field.

This is called the **Fermi contact interaction** since it depends on $|\psi_{ns}(0)|^2$ being finite. It can also be expressed as

$$H_{\text{HFS}} = A \mathbf{I} \cdot \mathbf{J} \quad (6.8)$$

because $\mathbf{J} = \mathbf{s}$ for $l = 0$. It is useful to write down this more general form at an early stage since it turns out that an interaction proportional to $\mathbf{I} \cdot \mathbf{J}$ is also obtained when $l \neq 0$.

We have already considered the effect of an interaction proportional to a dot product of two angular momenta when looking at the spin-orbit interaction $\beta \mathbf{S} \cdot \mathbf{L}$ (eqn 5.4). In the same way the hyperfine interaction in eqn 6.8 causes $\mathbf{I}$ and $\mathbf{J}$ to change direction but the total angular momentum of the atom $\mathbf{F} = \mathbf{I} + \mathbf{J}$ remains constant. The quantities $\mathbf{I} \cdot \mathbf{B}$ and $\mathbf{J} \cdot \mathbf{B}$ are not constant in this precession of $\mathbf{I}$ and $\mathbf{J}$ around $\mathbf{F}$. Therefore M_I and M_J are not good quantum numbers; we use F and M_F instead and the eigenstates of H_{HFS} are $|IJFM_F\rangle$.⁵ The expectation value of eqn 6.8 gives

$$E_{\text{HFS}} = A \langle \mathbf{I} \cdot \mathbf{J} \rangle = \frac{A}{2} \{F(F+1) - I(I+1) - J(J+1)\}. \quad (6.9)$$

Example 6.1 *Hyperfine structure of the $1s \ ^2S_{1/2}$ ground state of hydrogen*

The lowest level of the hydrogen atom is $1s \ ^2S_{1/2}$, i.e. $J = \frac{1}{2}$ and the proton has spin $I = 1/2$ so that $F = 0$ and 1 and these hyperfine levels have energies

$$E_{\text{HFS}} = \frac{A}{2} \{F(F+1) - I(I+1) - J(J+1)\} = \begin{cases} A/4 & \text{for } F = 1, \\ -3A/4 & \text{for } F = 0. \end{cases}$$

The splitting between the hyperfine levels is $\Delta E_{\text{HFS}} = A$ (see Fig. 6.3). Substituting for $|\psi_{ns}(0)|^2$ from eqn 2.22 gives

$$A = \frac{2}{3} \mu_0 g_s \mu_B g_I \mu_N \frac{Z^3}{\pi a_0^3 n^3}. \quad (6.10)$$

⁵This should be compared with the *LS*-coupling scheme where combinations of the eigenstates $|LM_LSM_S\rangle$ form eigenstates of the spin-orbit interaction $|LSJM_J\rangle$. The same warning issued for *LS*-coupling and the spin-orbit interaction $\beta \mathbf{S} \cdot \mathbf{L}$ also applies. It is important not to confuse *IJ*-coupling and the interaction $A \mathbf{I} \cdot \mathbf{J}$. In the *IJ*-coupling scheme I and J are good quantum numbers and the states are $|IJM_IM_J\rangle$ or $|IJFM_F\rangle$. The latter are eigenstates of the interaction $A \mathbf{I} \cdot \mathbf{J}$.

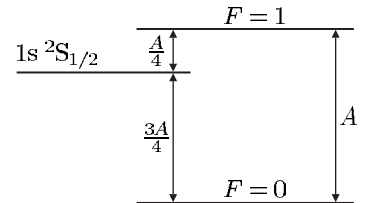


Fig. 6.3 The splitting between the hyperfine levels in the ground state of hydrogen.

The proton has a g -factor of $g_I = 5.6$, so for $n = 1 = Z$ we have

$$\frac{\Delta E_{\text{HFS}}}{h} = 1.42 \text{ GHz}.$$

⁶This hyperfine transition in hydrogen is the one detected in radio astronomy, where it is commonly referred to by its wavelength as the 21 cm line.

This is very close to the measured value of $1\,420\,405\,751.7667 \pm 0.0009 \text{ Hz}$. This frequency has been measured to such extremely high precision because it forms the basis of the hydrogen maser, described below.⁶

6.1.2 Hydrogen maser

Maser stands for microwave amplification by stimulated emission of radiation and such devices were the precursors of lasers, although nowadays they are less widespread than the devices using light. Actually, lasers are usually operated as oscillators rather than amplifiers but the acronym with ‘o’ instead of ‘a’ is not so good. A schematic of a hydrogen maser is shown in Fig. 6.4; it operates in the following way.

- Molecular hydrogen is dissociated in an electrical discharge.
- Atoms effuse from the source to form a beam in an evacuated chamber.
- The atoms pass through a region with a strong magnetic field gradient (from a hexapole magnet) that focuses atoms in the upper hyperfine level ($F = 1$) into a glass bulb. The atomic beam contains atoms in both hyperfine levels but the state selection by the magnet⁷ leads to a

⁷This works on the same basic principle as the Stern–Gerlach experiment in Section 6.4.

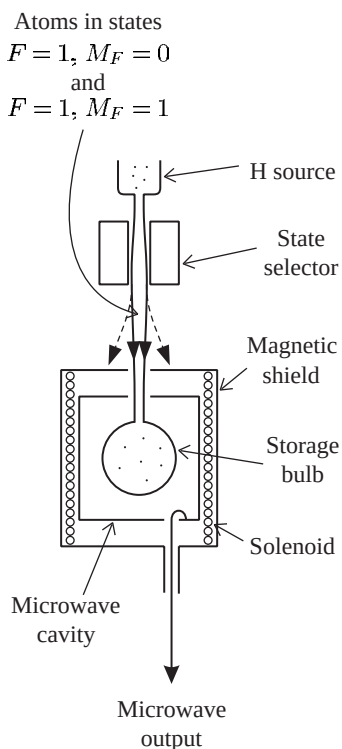


Fig. 6.4 The hydrogen maser. The principle of operation is described in the text. A magnetic shield excludes external fields and the solenoid creates a small stable magnetic field. In this way the frequency shift produced by the Zeeman effect on the hyperfine structure is controlled (in the same way as in the atomic clock described in Section 6.4.2).

population inversion in the bulb, i.e. the population in $F = 1$ exceeds that in $F = 0$; this gives more stimulated emission than absorption and hence gain, or amplification of radiation at the frequency of the transition.

- The atoms bounce around inside the bulb—the walls have a ‘non-stick’ coating of teflon so that collisions do not change the hyperfine level.
- The surrounding microwave cavity is tuned to the 1.42 GHz hyperfine frequency and maser action occurs when there are a sufficient number of atoms in the upper level. Power builds up in a microwave cavity, some of which is coupled out through a hole in the wall of the cavity.

The maser frequency is very stable—much better than any quartz crystal used in watches. However, the output frequency is not precisely equal to the hyperfine frequency of the hydrogen atoms because of the collisions with the walls (see Section 6.4).

6.1.3 Hyperfine structure for $l \neq 0$

Electrons with $l \neq 0$, orbiting around the nucleus, give a magnetic field

$$\mathbf{B}_e = \frac{\mu_0}{4\pi} \left\{ \frac{-e\mathbf{v} \times (-\mathbf{r})}{r^3} - \frac{\boldsymbol{\mu}_e - 3(\boldsymbol{\mu}_e \cdot \hat{\mathbf{r}})\hat{\mathbf{r}}}{r^3} \right\}, \quad (6.11)$$

where $-\mathbf{r}$ gives the position of the nucleus with respect to the orbiting electron. The first term arises from the orbital motion.⁸ It contains the cross-product of $-e\mathbf{v}$ with $-\mathbf{r}$, the position vector of the nucleus relative to the electron, and $-e\mathbf{r} \times \mathbf{v} = -2\mu_B \mathbf{l}$. The second term is just the magnetic field produced by the spin dipole moment of the electron $\boldsymbol{\mu}_e = -2\mu_B \mathbf{s}$ (taking $g_s = 2$) at a position $-\mathbf{r}$ with respect to the dipole.⁹ Thus we can write

$$\mathbf{B}_e = -2\frac{\mu_0}{4\pi} \frac{\mu_B}{r^3} \left\{ \mathbf{l} + \frac{3(\mathbf{s} \cdot \mathbf{r})\mathbf{r}}{r^2} \right\}. \quad (6.12)$$

This combination of orbital and spin-dipolar fields has a complicated vector form.¹⁰ However, we can again use the argument (in Section 5.1) that in the vector model there is rapid precession around $\mathbf{J}$, and any components perpendicular to this quantisation axis average to zero, so that only components along $\mathbf{J}$ have a non-zero time-averaged value.¹¹ The projection factor can be evaluated exactly (Woodgate 1980) but we shall assume that it is approximately unity giving

$$\mathbf{B}_e \sim -2\frac{\mu_0}{4\pi} \frac{\mu_B}{r^3} \langle \mathbf{l} + \frac{3(\mathbf{s} \cdot \mathbf{r})\mathbf{r}}{r^2} \rangle \cdot \mathbf{J}. \quad (6.13)$$

Thus from eqns 6.1 and 6.3 we find that the hyperfine interaction for electrons with $l \neq 0$ has the same form $A\mathbf{I} \cdot \mathbf{J}$ as eqn 6.8. This form of interaction leads to the following interval rule for hyperfine structure:

$$E_F - E_{F-1} = AF. \quad (6.14)$$

This interval rule is derived in the same way as eqn 5.8 for fine structure but with I , J and F instead of L , S and J , as shown in Exercise 6.5.

⁸This resembles the Biot–Savart law of electromagnetism:

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I \mathbf{ds} \times \mathbf{r}}{r^3}.$$

Roughly speaking, the displacement along the direction of the current is related to the electron’s velocity by $\mathbf{ds} = \mathbf{v} dt$, where dt is a small increment of time, and the current is related to the charge by $I dt = Q$.

The spin–orbit interaction can be very crudely ‘justified’ in a similar way, by saying that the electron ‘sees’ the nucleus of charge $+Ze$ moving round it; for a hydrogenic system this simplistic argument gives

$$\mathbf{B}_{\text{orbital}} = -2Z\frac{\mu_0}{4\pi} \frac{\mu_B}{r^3} \mathbf{l}.$$

The Thomas precession factor does not occur in hyperfine structure because the frame of reference is not rotating.

⁹See Blundell (2001).

¹⁰The same two contributions to the field also occur in the fine structure of helium; the field produced at the position of one electron by the orbital motion of the other electron is called the spin–other-orbit interaction, and a spin–spin interaction arises from the field produced by the magnetic dipole of one electron at the other electron.

¹¹In quantum mechanics this corresponds to saying that the matrix elements of any vector operator in the eigenbasis $|JM_J\rangle$ are proportional to $\mathbf{J}$, i.e. $\langle JM_J|\mathbf{B}_e|JM_J\rangle \propto \langle JM_J|\mathbf{J}|JM_J\rangle$. This is a consequence of the Wigner–Eckart theorem that was mentioned in Section 5.1.

Table 6.1 Comparison of fine and hyperfine structures.

	Fine structure in the <i>LS</i> -coupling scheme	Hyperfine structure in the <i>IJ</i> -coupling scheme
Interaction	$\beta \mathbf{L} \cdot \mathbf{S}$	$A \mathbf{I} \cdot \mathbf{J}$
Total angular momentum	$\mathbf{J} = \mathbf{L} + \mathbf{S}$	$\mathbf{F} = \mathbf{I} + \mathbf{J}$
Eigenstates	$ LSJM_J\rangle$	$ IJFM_F\rangle$
Energy, E	$\frac{\beta}{2} \{J(J+1) - L(L+1) - S(S+1)\}$	$\frac{A}{2} \{F(F+1) - I(I+1) - J(J+1)\}$
Interval rule	$E_J - E_{J-1} = \beta J$ (if $E_{s-o} < E_{re}$)	$E_F - E_{F-1} = AF$ (if $A \gg \Delta E_{\text{Quadrupole}}$)

¹²This interval rule for magnetic dipole hyperfine structure can be disrupted by the quadrupole interaction. Some nuclei are not spherical and their charge distribution has a quadrupole moment that interacts with the gradient of the electric field at the nucleus. This electric quadrupole interaction turns out to have an energy comparable to the interaction of the magnetic dipole moment μ_I with $\mathbf{B}_e$. Nuclei, and atoms, do not have static electric dipole moments (for states of definite parity).

That exercise also shows how this rule can be used to deduce F and hence the nuclear spin I from a given hyperfine structure.¹²

The hyperfine-structure constant $A(n, l, j)$ is smaller for $l > 0$ than for $l = 0$ and the same n . Exact calculation shows that the hyperfine-structure constants of the hydrogenic levels $np \ ^2P_{1/2}$ and $ns \ ^2S_{1/2}$ are in the ratio

$$\frac{A(n \ ^2P_{1/2})}{A(n \ ^2S_{1/2})} = \frac{1}{3}. \quad (6.15)$$

This ratio is smaller in the alkalis, e.g. $\sim 1/10$ in the examples below, because the closed shells of electrons screen the nuclear charge more effectively for p-electrons than for s-electrons.

6.1.4 Comparison of hyperfine and fine structures

The analogy between hyperfine and fine structures is summarised in Table 6.1.

For fine structure in the alkalis we found the Landé formula (eqn 4.13)

$$E_{\text{FS}} \sim \frac{Z_i^2 Z_o^2}{(n^*)^3} \alpha^2 hc R_\infty. \quad (6.16)$$

The Z^4 scaling for a hydrogenic system is reduced to $E_{\text{FS}} \propto Z^2$ for neutral atoms since the effective outer atomic number is $Z_o = 1$, and $Z_i \sim Z$ gives a reasonable approximation in the inner region. Applying similar considerations to the hyperfine structure shows that the dependence on Z^3 in eqn 6.10 reduces to

$$E_{\text{HFS}} \sim \frac{Z_i Z_o^2}{(n^*)^3} \frac{m_e}{M_p} \alpha^2 hc R_\infty. \quad (6.17)$$

The mass ratio arises from $\mu_N/\mu_B = m_e/M_p$. Hyperfine structure scales as Z , whereas fine structure scales as Z^2 ; thus E_{HFS} varies much less than E_{FS} , as the following comparison of the splittings for Na and Cs shows.

Na, $Z = 11$	Cs, $Z = 55$
$E(3p\ ^2P_{3/2}) - E(3p\ ^2P_{1/2}),$ $\Delta f_{\text{FS}} = 510\text{ GHz}$	$E(6p\ ^2P_{3/2}) - E(6p\ ^2P_{1/2}),$ $\Delta f_{\text{FS}} = 16\,600\text{ GHz}$
For the ground state $3s\ ^2S_{1/2},$ $\Delta f_{\text{HFS}} = 1.8\text{ GHz}$	For the ground state $6s\ ^2S_{1/2},$ $\Delta f_{\text{HFS}} = 9.2\text{ GHz}$
For $3p\ ^2P_{1/2},$ $\Delta f_{\text{HFS}} = 0.18\text{ GHz}$	For $6p\ ^2P_{1/2},$ $\Delta f_{\text{HFS}} = 1.2\text{ GHz}$

The hyperfine splitting of the ground states and the fine-structure splitting of the first excited states are indicated on the plot of energies against Z in Fig. 6.5. The values shown are only a guideline; e.g. different

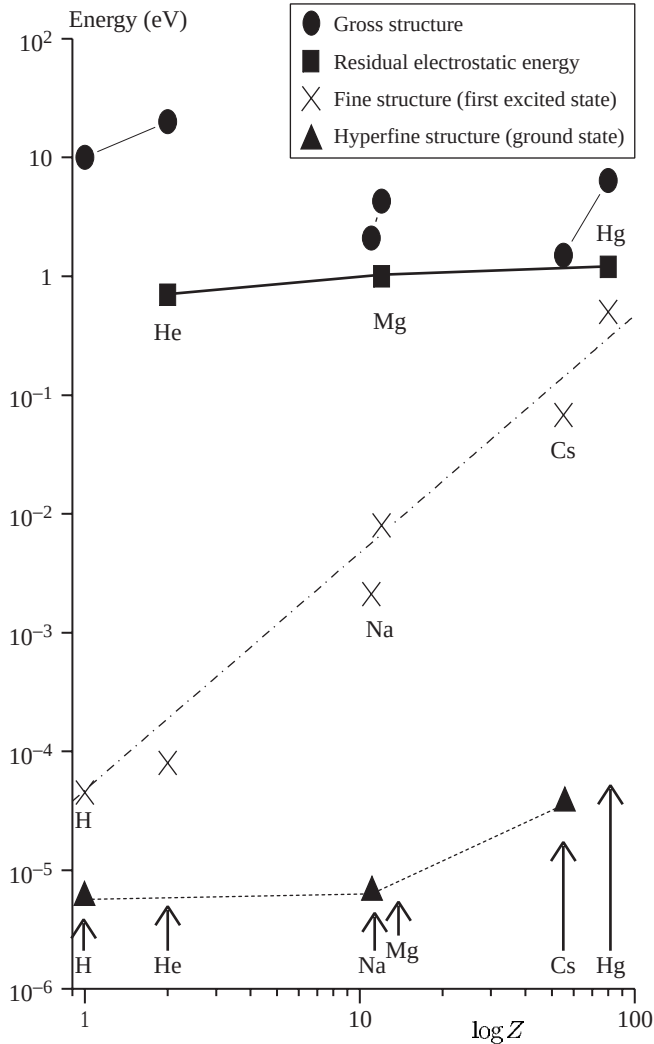


Fig. 6.5 A logarithmic plot of the energy of various structures against atomic number Z ; the hyperfine splitting of the ground state is plotted with data from Fig. 5.7. All the points are close to the maximum values of that quantity for low-lying configurations, terms, levels and hyperfine levels (as appropriate) of neutral atoms with one or two valence electrons, and these illustrate how these quantities vary with Z . This is only a rough guideline in particular cases; higher-lying configurations in neutral atoms have smaller values and in highly-ionized systems the structures have higher energies.

isotopes of the same element have different hyperfine splittings because the magnetic moment μ_I depends on the nuclear structure. The ground state of hydrogen has an especially large hyperfine structure that is greater than that of lithium ($Z = 3$), see Exercise 6.3.

Example 6.2 *Hyperfine structure of europium*

Figure 6.6 shows an experimental trace of a $4f^7 6s^2 \ ^8S_{7/2} - 4f^7 6s 6d \ ^8D_{11/2}$ transition in europium obtained by Doppler-free laser spectroscopy (Kronfeldt and Weber 1991).¹³ The ground level ($4f^7 6s^2 \ ^8S_{7/2}$) has a small hyperfine structure, from the unpaired f-electrons, that causes the small splitting of the peaks labelled 3, 4, 5, 6 and 7 (barely resolved for peak 3); however, this detail will not be considered further in the following analysis,¹⁴ which concentrates on the much larger hyperfine structure of the $4f^7 6s 6d \ ^8D_{11/2}$ level that arises mainly from the unpaired s-electron. The spectrum has a dozen main peaks. Since $J = 11/2$ a naive analysis might suppose that $I \geq J$ and so the observed peaks arise from transitions to the $2J + 1 = 12$ hyperfine levels expected in this case, i.e. $F = I + J, I + J - 1, \dots, I - J + 1$ and $I - J$. This is obviously wrong for various reasons: it is clear that the pattern of all twelve peaks does not fit any simple rule and also this element has two stable isotopes ^{151}Eu and ^{153}Eu . As indicated by their similar shape, the peaks labelled 3, 4, 5, 6 and 7 all belong to same isotope (^{151}Eu) and this can be verified by the interval rule as shown in the following table.

I Peak	II Position (GHz)	III $\frac{E_{F-1} - E_F}{h}$ (GHz)	IV Ratio of differences, x	V $\frac{x}{x-1}$
3	21.96	—	—	—
4	19.14	2.82	—	—
5	15.61	3.53	1.252	5.0
6	11.37	4.24	1.201	6.0
7	6.42	4.95	1.167	7.0
g	0.77	5.65	1.141	8.1

¹³Two-photon spectroscopy is explained in Section 8.4

¹⁴It is straightforward to apply the interval rule even when both the lower and upper levels have hyperfine structure.

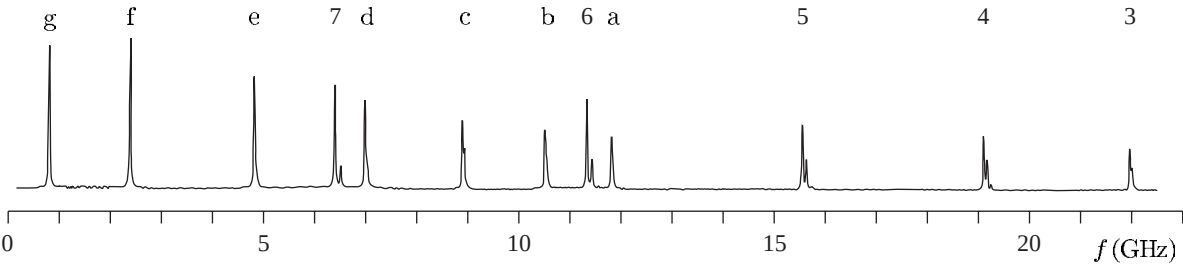


Fig. 6.6 An experimental trace of a $4f^7 6s^2 \ ^8S_{7/2} - 4f^7 6s 6d \ ^8D_{11/2}$ transition in europium obtained by Doppler-free laser spectroscopy (Kronfeldt and Weber 1991). Copyright 1991 by the American Physical Society.

Column II in this table gives the positions of the peaks¹⁵ measured from Fig. 6.6. Column III gives the difference between the frequencies in column II (the intervals between the peaks), e.g. $21.96 - 19.14 = 2.82$. Column IV gives the ratio of the intervals in column III, e.g. $3.53/2.82 = 1.252$. The interval rule for hyperfine structure in eqn 6.14 predicts that

$$x = \frac{E_F - E_{F-1}}{E_{F-1} - E_{F-2}} = \frac{AF}{A(F-1)} = \frac{F}{F-1}. \quad (6.18)$$

Rearrangement gives the total angular momentum F in terms of x as

$$F = \frac{x}{x-1}. \quad (6.19)$$

The numerical values of this quantity in column V (that have been calculated from the data by the above procedure) confirm that F has the value used to label the peaks. Moreover, we find that peak g fits the interval rule with $F = 8$. Thus, since this level has $J = 11/2$, this isotope (¹⁵¹Eu) must have a nuclear spin of $I = 5/2$ —this follows from the rules for the addition of angular momentum which allow values of F between $F_{\max} = I + J = 8$ and $F_{\min} = |I - J| = 3$.¹⁶ Exercise 6.5 shows that the other six peaks a to f also obey an interval rule and they all belong to another isotope (¹⁵³Eu).

¹⁵The highest peak in the case of the closely-spaced pairs.

¹⁶The proof of this result using operators can be found in quantum mechanics texts. It can be justified by analogy with vector addition: the maximum value occurs when the two angular momentum vectors point in the same direction and the minimum value when they are anti-parallel.

6.2 Isotope shift

In addition to the (magnetic dipole) hyperfine interaction in eqn 6.8 there are several other effects that may have a comparable magnitude (or might even be larger).¹⁷ This section describes two effects that lead to a difference in the frequency of the spectral lines emitted by different isotopes of an element.

¹⁷The quadrupole interaction was noted in Table 6.1, but will not be discussed further.

6.2.1 Mass effects

In Chapter 1 we saw that, in the Bohr model, energies are proportional to the reduced mass of the electron, given in eqn 1.13, and this scaling also applies to the solutions of the Schrödinger equation. Thus a transition between two levels of energies E_1 and E_2 has a wavenumber $\tilde{\nu} = (E_2 - E_1)/hc$ that is related to $\tilde{\nu}_\infty$, the value for a ‘theoretical’ atom with a nucleus of infinite mass, by

$$\tilde{\nu} = \tilde{\nu}_\infty \times \frac{M_N}{m_e + M_N}, \quad (6.20)$$

where M_N is the mass of the nucleus. However, $\tilde{\nu}_\infty$ cannot be measured. What we can observe is the difference in wavenumbers between two isotopes of an element, e.g. hydrogen and deuterium for $Z = 1$. In general, for two isotopes with atomic masses A' and A'' , we can make the approximation $M_N = A'M_p$ or $A''M_p$, so that¹⁸

¹⁸Strictly, atomic mass units should be used rather than M_p . The difference between the mass of an atom and its nucleus equals the mass of the electrons including the contribution from their binding energy. However, for this estimate we do not need to know M_N precisely.

$$\begin{aligned}
\Delta\tilde{\nu}_{\text{Mass}} &= \tilde{\nu}_{A''} - \tilde{\nu}_{A'} = \frac{\tilde{\nu}_{\infty}}{1 + m_e/A''M_p} - \frac{\tilde{\nu}_{\infty}}{1 + m_e/A'M_p} \\
&\simeq \tilde{\nu}_{\infty} \left\{ 1 - \frac{m_e}{A''M_p} - \left(1 - \frac{m_e}{A'M_p} \right) \right\} \\
&\simeq \frac{m_e}{M_p} \frac{\delta A}{A' A''} \tilde{\nu}_{\infty}.
\end{aligned} \tag{6.21}$$

This is called the **normal mass shift** and the energy difference $hc\Delta\tilde{\nu}_{\text{Mass}}$ is plotted in Fig. 6.7, assuming that $\delta A = 1$, $A \simeq A' \simeq 2Z$, and that $E_2 - E_1 \simeq 2\text{ eV}$ for a visible transition. The mass shift is largest for hydrogen and deuterium where $A'' = 2A' \simeq 2M_p$ (Exercise 1.1); it is larger than the fine structure in this case. For atoms with more than one electron there is also a **specific mass shift** that has the same order of magnitude as the normal mass effect, but is much harder to calculate.¹⁹ Equation 6.20 shows that the mass shift always leads to the heavier isotope having a higher wavenumber—by definition the reduced mass of the electron is less than m_e , and as the atomic mass increases the energy levels become closer to those of the theoretical atom with a nucleus of infinite mass.

¹⁹See Exercise 6.12 and also Woodgate (1980).

6.2.2 Volume shift

Although nuclei have radii which are small compared to the scale of electronic wavefunctions, $r_N \ll a_0$, the nuclear size has a measurable effect on spectral lines. This finite nuclear size effect can be calculated as a perturbation in two complementary ways. A simple method uses Gauss' theorem to determine how the electric field of the nuclear charge distribution differs from $-Ze/4\pi\epsilon_0 r^2$ for $r \lesssim r_N$ (see Woodgate 1980).

Alternatively, to calculate the electrostatic interaction of two overlapping charge distributions (as in eqn 3.15, for example) we can equally well find the energy of the nucleus in the potential created by the electronic charge distribution (in an analogous way to the calculation of the magnetic field at the nucleus created by s-electrons in Section 6.1.1). The charge distributions for an s-electron and a typical nucleus closely resemble those shown in Fig. 6.2.²⁰ In the region close to the nucleus there is a uniform electronic charge density

$$\rho_e = -e |\psi(0)|^2. \tag{6.22}$$

Using Gauss' theorem to find the electric field at the surface of a sphere of radius r in a region of uniform charge density shows that the electric field is proportional to r . Integration gives the electrostatic potential:

$$\phi_e(r) = -\frac{\rho_e r^2}{6\epsilon_0}. \tag{6.23}$$

The zero of the potential has been chosen to be $\phi_e(0) = 0$. Although this is not the usual convention the difference in energy that we calculate does not depend on this choice.²¹ With this convention a point-like nucleus

²⁰For hyperfine structure we were concerned with the nuclear magnetic moment arising from the constituent protons and neutrons. To calculate the electrostatic effect of a finite nuclear size, however, we need to consider the charge distribution of the nucleus, i.e. the distribution of protons. This has a shape that is similar to, but not the same as, the distribution of nuclear matter (see nuclear physics texts). The essential point for atomic structure is that all the nuclear distributions extend over a distance small compared to the electronic wavefunctions, as illustrated in Fig. 6.2.

²¹If this worries you, then put an arbitrary constant ϕ_0 in the equation, arising from the integration of the electric field to give the electrostatic potential, and show that the answer does not depend on ϕ_0 .

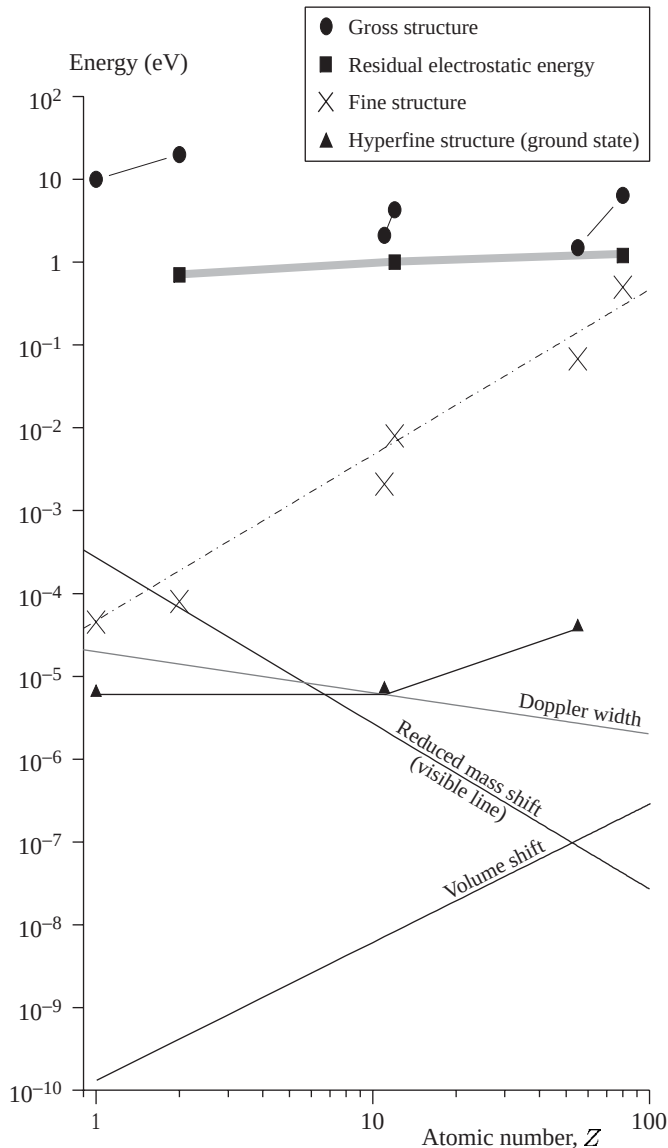


Fig. 6.7 A plot of the energy of various structures against atomic number Z . This includes the data shown on Fig. 6.5, but the scale has been changed to show the effects of lower energy. The contributions to the isotope shift from the mass and volume effects are plotted using eqns 6.21 and 6.26, respectively. Although these effects arise from completely independent physical effects, they have a similar magnitude for medium-heavy elements (and can be comparable to the magnetic dipole hyperfine structure of an excited state, which is considerably smaller than the ground-state splitting). The Doppler width for a visible transition with energy 2 eV is plotted to indicate the typical experimental limit in spectroscopic measurements using light; the line has a slope of $-1/2$ because $\Delta f_D \propto 1/\sqrt{M}$ and the atomic mass A is approximately $A \sim 2Z$, so that $\Delta f_D \propto Z^{-1/2}$ (see eqn 8.7). For hydrogen the Doppler width is almost equal to the fine-structure splitting (and much larger than hyperfine structure). The Doppler broadening of visible lines in sodium is comparable with its hyperfine structure; for heavier elements the ground-state hyperfine structure can be resolved in a vapour at room temperature. This gives an indication of the importance of Doppler-free techniques described in Chapter 8. The Zeeman effect of a magnetic flux density of $B = 1$ T is equivalent to a frequency shift of 14 GHz.

has zero potential energy, and for a distribution of nuclear charge $\rho_N(r)$ the potential energy is

$$\begin{aligned}
 E_{\text{Vol}} &= \iiint \rho_N \phi_e d^3\mathbf{r} = \frac{e}{6\epsilon_0} |\psi(0)|^2 \iiint \rho_N r^2 d^3\mathbf{r} \\
 &= \frac{Z e^2}{6\epsilon_0} |\psi(0)|^2 \langle r_N^2 \rangle.
 \end{aligned} \tag{6.24}$$

The integral gives the mean-square charge radius of the nucleus $\langle r_N^2 \rangle$ times the charge Ze . This volume effect only applies to configurations with s-electrons. The liquid drop model gives a formula for the radius

²²More accurate calculations can be made directly from eqn 6.24 in individual cases, e.g. the 1s configuration in hydrogen has

$$E_{\text{Vol}} = \frac{4}{3} \frac{\langle r_{\text{N}}^2 \rangle}{a_0^2} hc R_{\infty} \simeq 5 \times 10^{-9} \text{ eV}.$$

The proton has a root-mean-square charge radius of $\langle r_{\text{N}}^2 \rangle^{1/2} = 0.875 \text{ fm}$ (CODATA value).

²³The size of the nucleus $\langle r_{\text{N}}^2 \rangle$ generally increases with A following the trend in eqn 6.25 but there are exceptions, e.g. a nucleus that is particularly stable because it has closed shells of nucleons can be smaller than a lighter nucleus. Experimental measurements of isotope shifts, and the deduced values of the volume effect, are used to study such behaviour. (Similarly, for atoms, the shell structure makes inert gas atoms exceptionally small. More generally, the variation of atomic size with atomic mass is opposite to that of ionization energy—alkali atoms are larger than nearby atoms in the periodic table.)

of a nucleus as

$$r_{\text{N}} \simeq 1.2 \times A^{1/3} \text{ fm}. \quad (6.25)$$

Using this equation and making the same approximations for the wave-function squared as in hyperfine structure (eqn 6.17), we can write the isotope shift caused by the volume effect as

$$\Delta \tilde{\nu}_{\text{Vol}} = \frac{\Delta E_{\text{Vol}}}{hc} \simeq \frac{\langle r_{\text{N}}^2 \rangle}{a_0^2} \frac{\delta A}{A} \frac{Z^2}{(n^*)^3} R_{\infty}. \quad (6.26)$$

This has been used to plot ΔE_{Vol} as a function of Z in Fig. 6.7, assuming that $\delta A = 1$, $A \sim 2Z$ and $n^* \sim 2$.²²

This volume effect decreases the binding energy of a given atomic level with respect to that of a ‘theoretical’ atom with a point charge. The resulting change in the transition depends on whether the effect occurs in the upper or lower level (see Exercise 6.9).²³

6.2.3 Nuclear information from atoms

We have shown that the nucleus has an observable effect on atomic spectra. If hyperfine structure is observed then one immediately knows that the nucleus has spin and the number of hyperfine components sets a lower limit on I (Example 6.1). The values of F , and hence I , can be deduced by checking the interval rule, and the sum rule for relative intensities (similar to that for fine structure in Section 4.6.1). In principle, the magnetic moment of the nucleus μ_I can be deduced from the hyperfine-structure constant A , e.g. calculations such as that in Section 6.1.1 are accurate for light atoms. For atoms with a higher Z , the relativistic effects are important for the electronic wavefunction near the nucleus and it is more difficult to calculate $|\psi(0)|^2$. However, the electronic factors cancel in ratios of the hyperfine-structure constants of isotopes of the same element to give accurate ratios of their magnetic moments, i.e. if the μ_I is known for one isotope then it can be deduced for the other isotopes (see Exercise 6.4).

Similarly, isotope shifts give the difference in the nuclear sizes between isotopes, $\Delta \langle r_{\text{N}}^2 \rangle$, assuming that the mass effects are calculable.²⁴ To interpret this information, it is necessary to know the absolute value of the charge radius for one of the isotopes by another means, e.g. muonic X-rays. These transitions between the energy levels of a muon bound to an atomic nucleus have a very large volume effect, from which $\langle r_{\text{N}}^2 \rangle$ can be deduced (see Exercise 6.13).²⁵

²⁴See Woodgate (1980) for further details.

²⁵High-energy electron scattering experiments also probe the nuclear charge distribution.

6.3 Zeeman effect and hyperfine structure

The treatment of the Zeeman effect on hyperfine structure (in the IJ -coupling scheme) closely resembles that described in Section 5.5 for the LS -coupling scheme, and the detailed explanation of each step is not repeated here. The total atomic magnetic moment of the atom is the

sum of the electronic and nuclear moments (cf. eqn 5.9):

$$\boldsymbol{\mu}_{\text{atom}} = -g_J \mu_B \mathbf{J} + g_I \mu_N \mathbf{I} \simeq -g_J \mu_B \mathbf{J}. \quad (6.27)$$

Since $\mu_N \ll \mu_B$ we can neglect the nuclear contribution (for all but the most precise measurements), so that the Hamiltonian for the interaction with an external field $\mathbf{B}$ is just that for the electronic magnetic moment:

$$H = g_J \mu_B \mathbf{J} \cdot \mathbf{B}. \quad (6.28)$$

This interaction does not depend on the nuclear spin. However, its expectation value does depend on the hyperfine structure. We consider first the weak-field regime where the interaction with the external field is weaker than $A \mathbf{I} \cdot \mathbf{J}$ so that it can be treated as a perturbation to the hyperfine structure. We then treat the strong-field regime, and also the intermediate situation.

6.3.1 Zeeman effect of a weak field, $\mu_B B < A$

If the interaction with the external field in eqn 6.28 is weaker than the hyperfine interaction $A \mathbf{I} \cdot \mathbf{J}$, then in the vector model $\mathbf{J}$ and $\mathbf{I}$ move rapidly about their resultant $\mathbf{F}$, as illustrated in Fig. 6.8, whilst $\mathbf{F}$ itself precesses more slowly about the magnetic field (z -axis). In this regime F and M_F are good quantum numbers, but M_I and M_J are not. Taking the projection of the magnetic moments along $\mathbf{F}$ gives the effective Hamiltonian

$$H = g_J \mu_B \frac{\langle \mathbf{J} \cdot \mathbf{F} \rangle}{F(F+1)} \mathbf{F} \cdot \mathbf{B} = g_F \mu_B \mathbf{F} \cdot \mathbf{B} = g_F \mu_B B F_z, \quad (6.29)$$

where

$$g_F = \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} g_J. \quad (6.30)$$

Here the factor g_F arises from the projection of $\mathbf{J}$ onto $\mathbf{F}$, as illustrated in Fig. 6.9, in the just same way as g_J is given by the projection of $\mathbf{L}$ and $\mathbf{S}$ onto $\mathbf{J}$ in Section 5.5. The Zeeman energy is

$$E = g_F \mu_B B M_F. \quad (6.31)$$

As an example, consider the ground-state hyperfine levels in hydrogen ($I = J = 1/2$ and $g_J = g_s \simeq 2$). For $F = 1$ we find $g_F = 1$ so the three states $M_F = -1, 0$ and 1 are spaced by $\mu_B B$. The $F = 0, M_F = 0$ state has no first-order Zeeman shift (see Fig. 6.10).

In summary, the calculation of the Zeeman effect of a weak magnetic field on the hyperfine structure is simple because only the magnetic moment of the electron(s) along $\mathbf{J}$ contributes, whereas in the LS -coupling scheme there are components along both $\mathbf{L}$ and $\mathbf{S}$. However, I affects g_F because the nuclear angular momentum I is not small, and has a major effect on the IJF -triangle (thinking in terms of vectors as in Fig. 6.9), even though the nuclear magnetic moment is negligible.

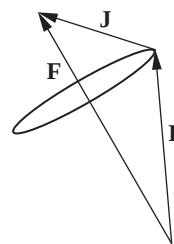


Fig. 6.8 The IJ -coupling scheme.

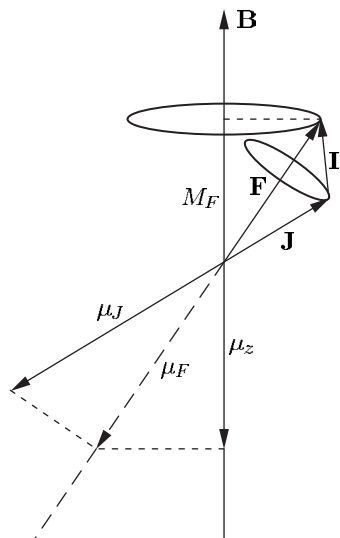


Fig. 6.9 The projection of the contributions to the magnetic moment from the atomic electrons along $\mathbf{F}$. The magnetic moment of the nucleus is negligible in comparison.

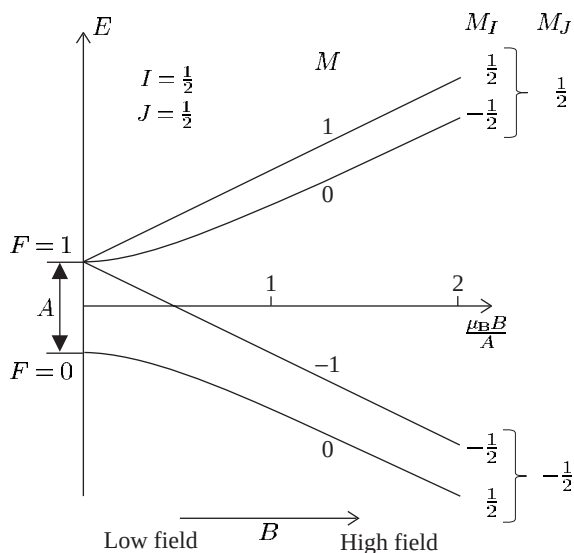


Fig. 6.10 The Zeeman effect on the hyperfine structure of the ground level of hydrogen $1s\ ^2S_{1/2}$. The interval between the $F = 0$ and $F = 1$ levels is A , as shown in Fig. 6.3. The zero of the energy scale has been chosen to be midway between the levels at zero field, which is convenient for the calculations in the text. (The two states with $M_F = 0$ in the low-field regime are mixed by the perturbation and then move apart as the magnetic flux density increases.) The quantity $x = \mu_B B/A$ is plotted on the horizontal axis and the low- and high-field regimes correspond to $x \ll 1$ and $x \gg 1$.

6.3.2 Zeeman effect of a strong field, $\mu_B B > A$

A strong field is one where the interaction with the external field, in eqn 6.28, is greater than $A \mathbf{I} \cdot \mathbf{J}$. This can readily be achieved for hyperfine structures since the Zeeman energy $\mu_B B$ in a field of 1 T is about 6×10^{-5} eV (corresponding to a frequency of 14 GHz), which is greater than all but the largest hyperfine structures in the ground configurations of heavy elements (see Fig. 6.7).²⁶ In this regime F is not a

²⁶But smaller than the fine structure of the first excited state, except for the lightest elements. High-lying levels, however, have smaller fine structure.

good quantum number and $\mathbf{J}$ precesses about $\mathbf{B}$.²⁷ The effect of the hyperfine interaction can be calculated as a perturbation on the $|IM_I JM_J\rangle$ eigenstates, i.e.

$$E_{\text{ZE}} = g_J \mu_B B M_J + \langle IM_I JM_J | A \mathbf{I} \cdot \mathbf{J} | IM_I JM_J \rangle \quad (6.32)$$

$$= g_J \mu_B B M_J + A M_I M_J. \quad (6.33)$$

The first term is the same as eqn 5.11. In the second term, $\mathbf{I} \cdot \mathbf{J} = I_x J_x + I_y J_y + I_z J_z$ and the x - and y -components average to zero in the precession about the field along the z -direction.²⁸

An example of the energy levels in a strong field is shown in Fig. 6.10 for the hydrogen ground state. The two energy levels with $M_J = \pm 1/2$ are both split into sub-levels with $M_I = \pm 1/2$ by the hyperfine interaction; eqn 6.33 shows that these sub-levels have a separation of $A/2$ (independent of the field strength).

6.3.3 Intermediate field strength

In Fig. 6.10 the low- and high-field energy levels follow the rule that two states never cross if they have the same value of M , where at low fields $M = M_F$ and at high fields $M = M_I + M_J$. This implies that

$$\begin{array}{ccc} & M_J & M_I \\ F = 1, & M_F = 0 & \rightarrow +1/2, -1/2, \\ F = 0, & M_F = 0 & \rightarrow -1/2, +1/2. \end{array}$$

This rule can be justified by showing that the operator $I_z + J_z$ commutes with all the interactions and it allows unambiguous connection of states even in more complex cases.²⁹ For the simple case of hydrogen the energy levels can be calculated at all fields by simple perturbation theory, as shown below.

Example 6.3 *The Zeeman effect on the hyperfine structure of hydrogen for all field strengths*

Figure 6.10 shows the energy levels for all field strengths. The Zeeman energies of the $M = \pm 1$ states are $\pm \mu_B B$ for all fields because their wavefunctions are not mixed ($g_F = 1$ from eqn 6.30). The $M_F = 0$ states have no first-order shift but the magnetic field mixes these two states in the $F = 0$ and 1 hyperfine levels; the matrix element between them is

$$-\langle F = 1, M_F = 0 | \boldsymbol{\mu} \cdot \mathbf{B} | F = 0, M_F = 0 \rangle = \zeta \mu_B B.$$

Such (off-diagonal) matrix elements can be evaluated by angular momentum theory, but in this simple case we can get by without using Clebsch–Gordan coefficients (leaving ζ as an undetermined constant for the time being). The Hamiltonian for the two-level system is

$$H = \begin{pmatrix} A/2 & \zeta \mu_B B \\ \zeta \mu_B B & -A/2 \end{pmatrix}. \quad (6.34)$$

²⁷The nuclear angular momentum $\mathbf{I}$ does not precess around $\mathbf{B}$ because $-\boldsymbol{\mu}_I \cdot \mathbf{B}$ is negligible. In this regime the interaction $A \mathbf{I} \cdot \mathbf{J}$ makes $\mathbf{I}$ precess about the mean direction of $\mathbf{J}$, which is parallel to $\mathbf{B}$. Thus effectively $\mathbf{I}$ precesses about the axis defined by $\mathbf{B}$ (but not because of $-\boldsymbol{\mu}_I \cdot \mathbf{B}$). The vector model picture requires careful thought because of the subtle differences from the Paschen–Back effect (Fig. 5.14). In the quantum mechanical description this is taken into account by considering the relative magnitudes of the perturbations: $|\boldsymbol{\mu}_e \cdot \mathbf{B}| > |A \mathbf{I} \cdot \mathbf{J}| > |\boldsymbol{\mu}_I \cdot \mathbf{B}|$, where $\boldsymbol{\mu}_e$ is the magnetic moment of the atomic electrons in eqn 5.9.

²⁸This can be shown rigorously using the ladder operators

$$I_+ \equiv I_x + i I_y,$$

$$I_- \equiv I_x - i I_y,$$

and similarly for J_+ and J_- . These ladder operators change the magnetic quantum numbers, e.g.

$$I_+ |I M_I\rangle \propto |I M_I + 1\rangle.$$

Since

$$I_x J_x + I_y J_y = \frac{1}{2} (I_+ J_- + I_- J_+),$$

the expectation value of this part of $\mathbf{I} \cdot \mathbf{J}$ is zero (for states of given M_J and M_I as in eqn 6.32).

²⁹At low fields, $I_z + J_z \equiv F_z$, which clearly commutes with the interaction in eqn 6.29. At high fields the relevant interactions are proportional to J_z and $I_x J_x + I_y J_y + I_z J_z$, both of which commute with $I_z + J_z$.

³⁰Any choice for the point at which $E = 0$ leads to the same result, e.g. taking the unperturbed energies as $A/4$ and $-3A/4$ as in Fig. 6.3.

The energies are measured from the point midway between the hyperfine levels to streamline the algebra.³⁰ The energy eigenvalues are

$$E = \pm \sqrt{(A/2)^2 + (\zeta \mu_B B)^2}. \quad (6.35)$$

This exact solution for all fields is plotted in Fig. 6.10. The approximate solution for weak fields is

$$E_{\text{weak}} \simeq \pm \left\{ \frac{A}{2} + \frac{(\zeta \mu_B B)^2}{A} \right\}. \quad (6.36)$$

When $B = 0$ the two unperturbed levels have energies $\pm A/2$. The term proportional to B^2 is the usual second-order perturbation theory expression that causes the levels to *avoid* one another (hence the rule that states of the same M do not cross).³¹

³¹The avoided crossing of states that mix is a general feature of perturbation theory.

For strong fields, where $\mu_B B \gg A$, eqn 6.35 gives the energy of the $M = 0$ states as

$$E(F = \pm 1, M_F = 0) \simeq \pm \zeta \mu_B B. \quad (6.37)$$

In a strong field the energy levels of the system are given by $g_J \mu_B B M_J$ and the two $M_J = \pm 1/2$ states have Zeeman energies of $g \mu_B B M_J = \pm \mu_B B$. Comparison with eqn 6.37 shows that $\zeta = 1$, and so we have found the energies for all field strengths. The other two states have energies $E(M = \pm 1) = \frac{1}{2}A \pm \mu_B B$ for all values of B .

A similar approach can be used when $J = 1/2$ for arbitrary values of I , which applies to the ground states of the alkalis. For $I > 1/2$ there are more states to consider than in hydrogen (where $I = 1/2$), so the Hamiltonian will have larger dimensions than in eqn 6.34. Actually there were four basis states to consider in hydrogen, but because the perturbation mixes only two of them it was only necessary to diagonalise a 2×2 matrix.

6.4 Measurement of hyperfine structure

An apparatus similar to that for measuring the Zeeman effect, shown in Fig. 1.7(a), can be used to observe hyperfine structure—a magnet is not required because hyperfine structure arises from an internal magnetic field of the atom. Figure 6.11 shows a typical experimental trace obtained from such an experiment with a pressure-scanned Fabry–Perot étalon for the $5s5p\ ^3P_0$ – $5s6s\ ^3S_1$ line in cadmium. The 3P_0 level has no hyperfine structure (because $J = 0$) and the observed splitting comes entirely from the $5s6s\ ^3S_1$ level. Both s-electrons contribute to the field at the nucleus³² so this level has an exceptionally large hyperfine splitting that is greater than the Doppler broadening.

³²The two spins are aligned in a triplet state so the 5s- and 6s-electrons produce fields in the same direction.

The fact that the $J = 1$ level only gives two hyperfine levels implies that $I = 1/2$. If $I \geq 1$ then there would be three levels (by the rules for addition of angular momenta). One of these peaks might, however, be hidden underneath the large central peak? That this is not the case here

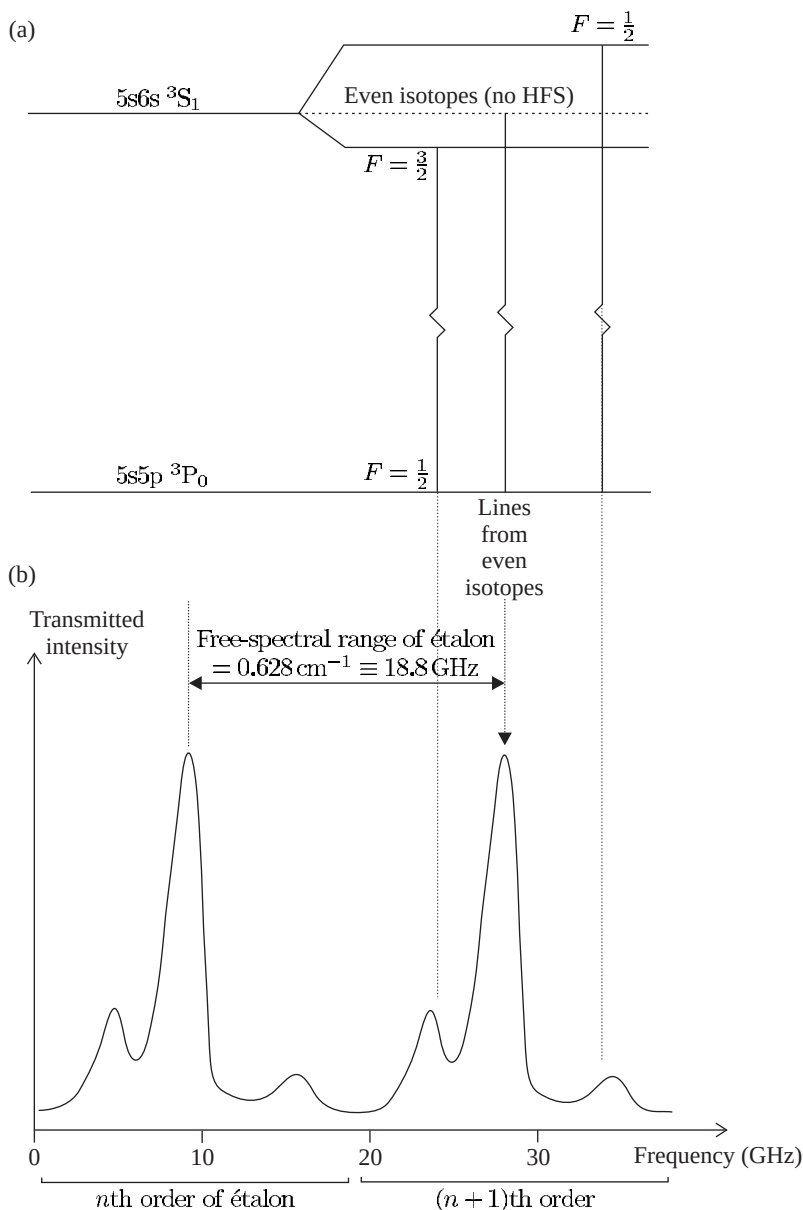


Fig. 6.11 (a) The $5s6s\ ^3S_1$ level in the odd isotopes of cadmium (^{111}Cd and ^{113}Cd) has a large hyperfine structure; the $F = 3/2$ hyperfine level lies below $F = 1/2$ because the nuclear g -factor is negative and similar in size for both odd isotopes. (The total magnetic field created by the unpaired s -electrons at the nucleus is anti-parallel to $\mathbf{J}$, as in the ground state of hydrogen and the alkalis.) (b) An experimental trace of the hyperfine structure for the $5s5p\ ^3P_0$ – $5s6s\ ^3S_1$ line at a wavelength of 468 nm obtained with a pressure-scanned Fabry–Perot étalon (as in Fig. 1.7). There are three peaks in each order of the étalon: a large peak from the isotopes with no nuclear spin $I = 0$ and hence no hyperfine structure (generally isotopes with an even number of nucleons so there are no unpaired spins within the nucleus); and two smaller peaks whose separation equals the hyperfine splitting of the $5s6s\ ^3S_1$ level. (Data from the Oxford Physics Teaching Laboratory practical course; further details are in Lewis (1977).)

can be checked by verifying that the observed peaks have the expected displacements from their centre of gravity (which is approximately at the position of the even isotopes).³³

It can be seen from Fig. 6.11 that optical spectroscopy is not generally suitable for measuring the Zeeman effect of hyperfine structure because spectral lines have a Doppler broadening that can be comparable with the hyperfine splitting. Thus in the low-field regime ($\mu_B B < A$) the Zeeman splitting is too small to be resolved. To show this quantitatively

³³This is not straightforward in this example because of the overlap of lines, but this could be done using a curve-fitting program on a computer. Additionally, there is a sum rule for the intensities as in fine structure.

we consider the Doppler width Δf_D of a line of wavelength λ :

$$\Delta f_D \simeq \frac{2u}{\lambda}, \quad (6.38)$$

where u is a typical atomic velocity. The factor of 2 appears because atoms can move towards and away from the observer. For this estimate we shall use the most probable velocity $u = \sqrt{2k_B T/m}$.³⁴ For cadmium at $T = 300\text{ K}$, we have $u \simeq 200\text{ m s}^{-1}$ and thus the Doppler width of the lines with a wavelength of $\lambda = 468\text{ nm}$ is $\Delta f_D = 2u/\lambda = 0.9\text{ GHz}$, whereas $A = 7.9\text{ GHz}$ for the $5s6s\ ^3S_1$ level. More generally, the Doppler width for a visible transition is plotted in Fig. 6.7 as a function of the atomic number. At room temperature the optical transitions of hydrogen, the lightest element, have a Doppler width slightly less than the fine-structure splitting of the first excited state, so in this case the low-field Zeeman effect cannot be observed even for fine structure (let alone hyperfine structure).³⁵ Figure 6.12 shows the results of an experimental observation of the hyperfine structure and isotope shift of tin by a technique of Doppler-free laser spectroscopy (that will be described in Section 8.2).

Doppler broadening is much less of a problem in direct measurements of the separation between hyperfine levels with microwave techniques (at frequencies of gigahertz), or the even smaller splitting of the Zeeman sub-levels that correspond to radio-frequency transitions. An example of a radio-frequency and microwave spectroscopy technique is outlined in the next section.

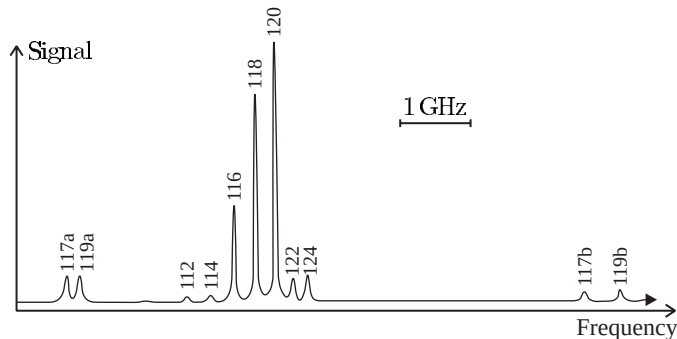
6.4.1 The atomic-beam technique

An atomic-beam apparatus can be understood as an extension of the original Stern–Gerlach apparatus illustrated in Fig. 6.13. In the original Stern–Gerlach experiment a beam of silver atoms was sent through a region of strong gradient of the magnetic field and further downstream the atoms were deposited on a glass plate. Upon inspecting the plate, Stern and Gerlach found that the atoms appeared in two distinct places, showing that the atomic beam was split into two directions. This told them that angular momentum is quantised and that it can have values

³⁴This gives a value of Δf_D that differs by only $\sqrt{\ln 2} = 0.8$ from the exact result for the full width at half maximum (FWHM) derived in Chapter 8.

³⁵The Zeeman splitting of a spectral line can only be resolved when the field is strong enough to give the Paschen–Back effect.

Fig. 6.12 Doppler-free laser spectroscopy of the $5p^2\ ^3P_0$ – $5p6s\ ^3P_1$ line of tin; the reduction in Doppler broadening (cf. Fig. 6.11) reveals the isotope shifts between the even isotopes, in addition to the hyperfine splitting of the odd isotopes. Each peak is labelled by the relative atomic mass. Each odd isotope gives rise to two peaks because of hyperfine structure (as in Fig. 6.11). For further details see Baird *et al.* (1983).



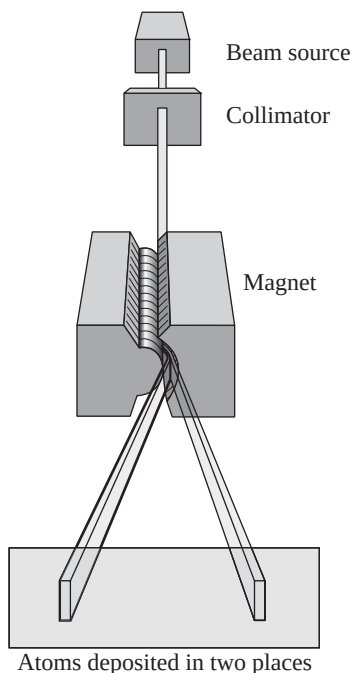


Fig. 6.13 In the original Stern–Gerlach apparatus a beam of silver atoms was sent through a region of strong gradient of the magnetic field and further downstream the atoms were deposited on a glass plate. This simple experiment had a great influence on the development of quantum ideas and provides a very useful illustration of the concepts of quantisation (Feynman *et al.* 1963–1965).

that are half-integer multiples of $\hbar$. (Orbital angular momentum would give $2l + 1$ components, with l an integer.) To interpret the experiment we consider an atom in a magnetic field $\mathbf{B}$ with a gradient along the z -axis.³⁶ The force exerted on the atom is (using eqn 5.11)

$$\text{Force} = -\frac{dE_{ZE}}{dz} = gJ\mu_B \frac{dB}{dz} M_J. \quad (6.39)$$

The ground level of silver, $5s\ ^2S_{1/2}$, has $l = 0$ and $J = s = 1/2$. Thus the force on atoms passing through the Stern–Gerlach magnet has the two values $F = \pm\mu_B dB/dz$ for $m_s = M_J = \pm\frac{1}{2}$. This explains the separation of the atomic beam into two components and provides direct observational evidence for quantisation since the two emerging beams are well defined. For a classical vector the z -component of angular momentum would be spread over a range between the maximum and minimum values.

An atomic-beam apparatus has two Stern–Gerlach-type magnets, as shown in Fig. 6.14, that create an inhomogeneous field in the A and B regions to deflect atoms according to their M_J state. The A and B regions have a strong field ($\mu_B B \gg A$), associated with the high gradient of magnetic field there, so M_J is a good quantum number and eqn 6.39 gives the force. The transition to be measured occurs in the C region between the two state-selecting magnets where, generally, the atoms are in the low-field regime. The principles of the operation of such an

³⁶Since $\nabla \cdot \mathbf{B} = 0$ we cannot have a gradient along the z -axis without there also being a gradient in another direction, but the effect of a gradient in a direction perpendicular to $\mathbf{B}$ averages to zero as the magnetic moment precesses around the z -axis—for a more rigorous discussion see Chapter 10.

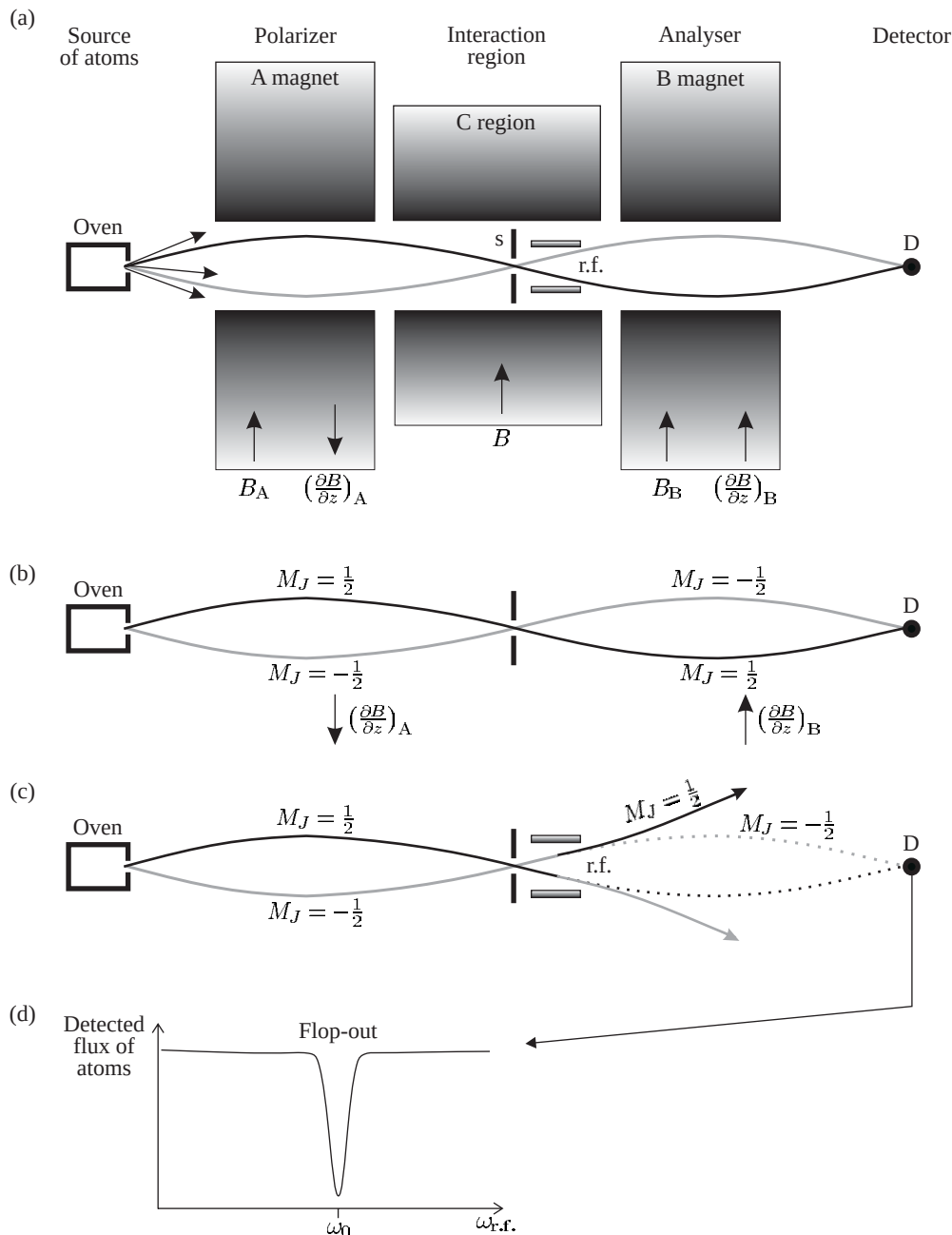


Fig. 6.14 (a) The magnetic resonance technique in an atomic beam. Atoms emerge from an oven and travel through the collimating slit 's' to the detector. The deflection of atoms by the magnetic field gradient in the A and B regions depends on M_J , as indicated. (b) Atoms that stay in the same M_J state are refocused onto the detector when the A and B regions have gradients in opposite directions. (c) Resonant interaction with radio-frequency radiation in the C region can change the M_J quantum number, $M_J = +\frac{1}{2} \leftrightarrow M_J = -\frac{1}{2}$, so that atoms no longer reach the detector. (In a real apparatus the C region may be up to several metres long.) This is known as the flop-out arrangement and gives a signal as in (d). Further details are given in the text and in Corney (2000).

atomic-beam experiment are as follows.

- Atoms emerge from an oven to form an atomic beam in an evacuated chamber. The atoms have a mean free path much greater than the length of the apparatus (i.e. there are no collisions).
- The deflection of atoms in the A and B regions depends on M_J . If these two regions have magnetic field gradients in the same direction, as indicated in Fig. 6.15, then atoms only reach the detector if their M_J quantum number changes in the C region, i.e. $M_J = +\frac{1}{2} \leftrightarrow M_J = -\frac{1}{2}$. This is known as the flop-in arrangement.³⁷
- As the atoms travel from A into the C region their state changes adiabatically to that in a low magnetic field as shown in Fig. 6.16 (and similarly as the field changes between the C and B regions). The transitions in the low-field region that can be observed are those that connect states with different values of M_J in the high-field regions, e.g. the transitions:

³⁷ The flop-out arrangement is shown in Fig. 6.14.

low frequency ($\Delta F = 0$):

$$F = 1, \quad M_F = 0 \leftrightarrow F = 1, \quad M_F = -1 \quad \text{with} \quad \Delta E = g_F \mu_B B;$$

higher frequency ($\Delta F = \pm 1$):

$$F = 0, \quad M_F = 0 \leftrightarrow F = 1, \quad M_F = 0 \quad \text{with} \quad \Delta E = A,$$

$$F = 0, \quad M_F = 0 \leftrightarrow F = 1, \quad M_F = 1 \quad \text{with} \quad \Delta E = A + g_F \mu_B B.$$

In this example, the $M_F = 1 \leftrightarrow M_F = 0$ change between the Zeeman sub-levels of the upper $F = 1$ level cannot be detected, but this

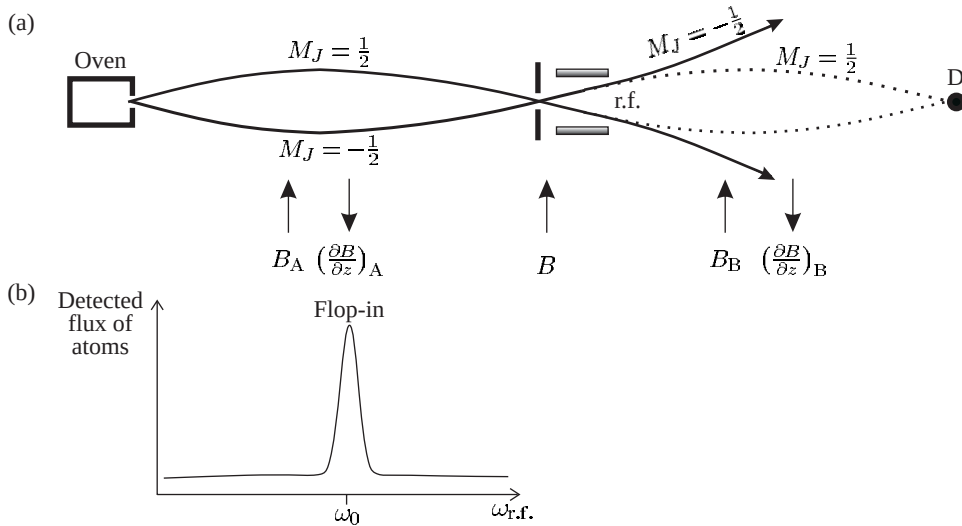
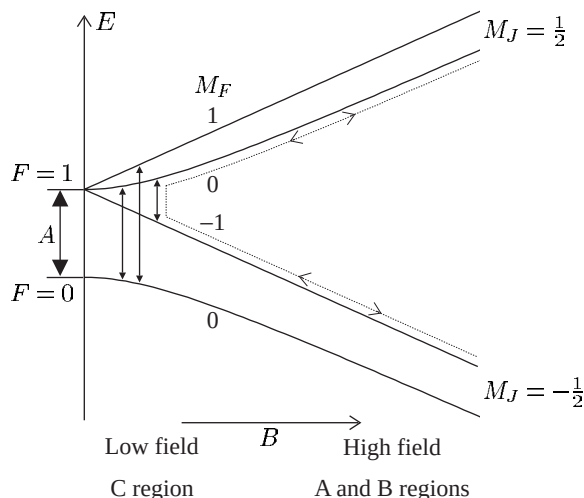


Fig. 6.15 (a) The trajectories of atoms in an atomic-beam apparatus similar to that shown in Fig. 6.14, but magnetic field gradients in the A and B regions have the same direction. Atoms only reach the detector if their M_J quantum number is changed in the C region by the interaction with radio-frequency radiation. This is known as the flop-in arrangement and gives a signal as in (b).

Fig. 6.16 The method of magnetic resonance in an atomic beam detects transitions that occur at low field (in the C region shown in Fig. 6.14) which cause a change in the quantum number M_J at high field (A and B regions). For example, an atom that follows the path indicated by the dotted line may start in the state $M_J = +\frac{1}{2}$, go adiabatically into the state $F = 1, M_F = 0$ in the C region, where it undergoes a transition to the state with $M_F = -1$ and then end up in the state $M_J = -\frac{1}{2}$ in the B region (or it may follow the same path in the opposite direction). A strong magnetic field gradient (which is associated with a high field) is required in the A and B regions to give an observable deflection of the atomic trajectories.



does not lead to any loss of information since the hyperfine-structure constant A and g_F can be deduced from the other transitions.

These transitions at microwave- and radio-frequencies are clearly not electric dipole transitions since they occur between sub-levels of the ground configuration and have $\Delta l = 0$ (and similarly for the maser). They are magnetic dipole transitions induced by the interaction of the oscillating magnetic field of the radiation with the magnetic dipole of the atoms. The selection rules for these M1 transitions are given in Appendix C.

6.4.2 Atomic clocks

An important application of the atomic-beam technique is atomic clocks, that are the primary standards of time. By international agreement the second is defined to be 9 192 631 770 oscillation periods of the hyperfine frequency in the ground state of ^{133}Cs (the only stable isotope of this element). Since all stable caesium atoms are identical, precise measurements of this atomic frequency in the National Standards Laboratories throughout the world should agree with each other, within experimental uncertainties.³⁸

The definition of the second is realised using the hyperfine frequency of the $F = 3, M_F = 0 \leftrightarrow F = 4, M_F = 0$ transition in caesium. This transition between two $M_F = 0$ states has no first-order Zeeman shift, but even the second-order shift has a significant effect at the level of precision required for a clock. The apparatus can be designed so that the dominant contribution to the line width comes from the finite interaction time τ as atoms pass through the apparatus (transit time). Fourier transform theory gives the line width as³⁹

$$\Delta f \sim \frac{1}{\tau} = \frac{v_{\text{beam}}}{l}, \quad (6.40)$$

³⁸Such quantum metrology has been used to define other fundamental constants, with the exception of the kilogram which is still defined in terms of a lump of platinum kept in a vault in Paris.

³⁹The next chapter gives a complete treatment of the interaction of atoms with radiation.

where v_{beam} is the typical velocity in the beam and l is the length of the interaction region. Therefore the atomic beams used as primary standards are made as long as possible. An interaction over a region 2 m long gives a line width of $\Delta f = 100 \text{ Hz}$.⁴⁰ Thus the quality factor is $f/\Delta f \sim 10^8$; however, the centre frequency of the line can be determined to a small fraction of the line width.⁴¹ As a result of many years of careful work at standards laboratories, atomic clocks have uncertainties of less than 1 part in 10^{14} . This illustrates the incredible precision of radio-frequency and microwave techniques, but the use of slow atoms gives even higher precision, as we shall see in Chapter 10.⁴² There is a great need for accurate timekeeping for the synchronisation of global telecommunications networks, and for navigation both on Earth through the global positioning system (GPS) and for satellites in deep space.

The atomic-beam technique has been described here because it furnishes a good example of the Zeeman effect on hyperfine structure, and it was also historically important in the development of atomic physics. The first atomic-beam experiments were carried out by Isador Rabi and he made numerous important discoveries. Using atomic hydrogen he showed that the proton has a magnetic moment of $2.8 \mu_N$, which was about three times greater than expected for a point-like particle (cf. the electron with μ_B , i.e. one unit of the relevant magnetic moment). This was the first evidence that the proton has internal structure. Other important techniques of radio-frequency spectroscopy such as optical pumping are described elsewhere, see Thorne *et al.* (1999) and Corney (2000).

Further reading

Further details of hyperfine structure and isotope shift including the electric quadrupole interaction can be found in Woodgate (1980). The discussion of magnetic resonance techniques in condensed matter (Blundell 2001) gives a useful complement to this chapter.

The classic reference on atomic beams is *Molecular beams* (Ramsey 1956). Further information on primary clocks can be found on the web sites of the National Physical Laboratory (for the UK), the National Institute of Standards and Technology (NIST, in the US), and similar sites for other countries. The two volumes by Vannier and Audouin (1989) give a comprehensive treatment of atomic clocks and frequency standards.

⁴⁰Caesium atoms have velocities of $v_{\text{Cs}} = (3k_B T / M_{\text{Cs}})^{1/2} = 210 \text{ m s}^{-1}$ at $T = 360 \text{ K}$.

⁴¹In a normal experiment it is hard to measure the centre of a line with an uncertainty of better than one-hundredth of the line width, i.e. a precision of only 1 in 10^{10} in this case.

⁴²The hydrogen maser achieves a long interaction time by confining the atoms in a glass bulb for $\tau \sim 0.1 \text{ s}$ to give a line width of order 10 Hz—the atoms bounce off the walls of the bulb without losing coherence. Thus masers can be more precise than atomic clocks but this does not mean that they are more accurate, i.e. the frequency of a given maser can be measured to more decimal places than that of an atomic-beam clock, but the maser frequency is shifted slightly from the hyperfine frequency of hydrogen by the effect of collisions with the walls. This shift leads to a frequency difference between masers that depends on how they were made. In contrast, caesium atomic clocks measure the unperturbed hyperfine frequency of the atoms. (The use of cold atoms improves the performance of both masers and atomic clocks—the above remarks apply to uncooled systems.)

Exercises

- (6.1) *The magnetic field in fine and hyperfine structure*
Calculate the magnetic flux density B at the centre of a hydrogen atom for the $1s\ ^2S_{1/2}$ level and also for $2s\ ^2S_{1/2}$.

Calculate the magnitude of the orbital magnetic field experienced by a $2p$ -electron in hydrogen (eqn 2.47).

- (6.2) *Hyperfine structure of lithium*

The figure shows the energy levels of lithium involved in the $2s$ – $2p$ transitions for the two isotopes ^6Li and ^7Li . (The figure is not to scale.)

Explain in simple terms why the hyperfine splitting is of order m_e/M_p smaller than the fine-structure splitting in the $2p$ configuration of lithium.

Explain, using the vector model or otherwise, why the hyperfine interaction splits a given J level into $2J + 1$ hyperfine levels if $J \leq I$ and $2I + 1$ levels if $I \leq J$. Hence, deduce the nuclear spin of ^6Li and give the values of the quantum numbers L , J and F for all its hyperfine levels. Verify that the

interval rule is obeyed in this case.

Determine from the data given the nuclear spin of ^7Li and give the values of L , J and F for each of the hyperfine levels on the figure. Calculate the hyperfine splitting of the interval marked X.

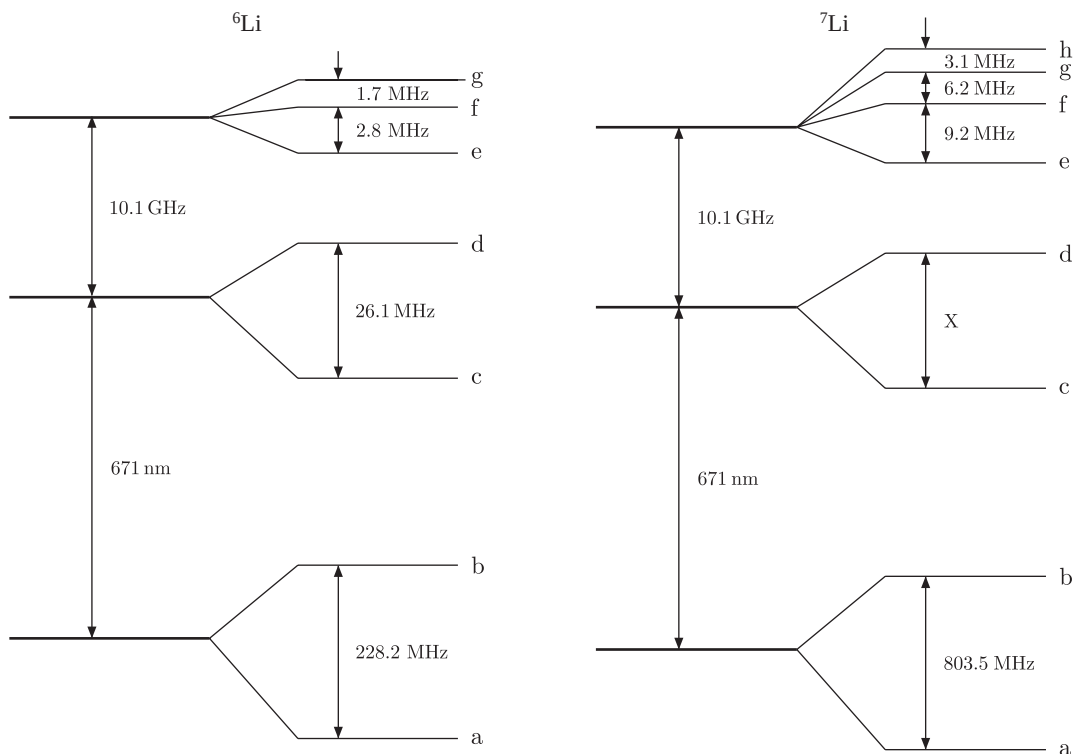
(For the hyperfine levels a to d the parameter A_{nlj} is positive for both isotopes.)

- (6.3) *Hyperfine structure of light elements*

Use the approximate formula in eqn 6.17 to estimate the hyperfine structure in the ground states of atomic hydrogen and lithium. Comment on the difference between your estimates and the actual values given for hydrogen in Section 6.1.1 and for Li in Exercise 6.2.

- (6.4) *Ratio of hyperfine splittings*

The spin and magnetic moment of the proton are $(1/2, 2.79\mu_N)$, of the deuteron $(1, 0.857\mu_N)$ and of ^3He $(1/2, -2.13\mu_N)$. Calculate the ratio of the ground-state hyperfine splittings of (a) atomic hydrogen and deuterium and (b) atomic hydrogen and the hydrogen-like ion $^3\text{He}^+$.



(6.5) Interval for hyperfine structure

- (a) Show that an interaction of the form $A \mathbf{I} \cdot \mathbf{J}$ leads to an interval rule, i.e. the splitting between two sub-levels is proportional to the total angular momentum quantum number F of the sub-level with the larger F .
- (b)

I Peak	II Position (GHz)
a	11.76
b	10.51
c	8.94
d	7.06
e	4.86
f	2.35

The table gives the positions of the six peaks in the spectrum shown in Fig. 6.6 that were not assigned quantum numbers in Example 6.2. It is the hyperfine structure of the upper level ($^8D_{11/2}$) of the transition in the isotope ^{153}Eu that determines the positions of these six peaks. What is the nuclear spin I of this isotope? Show that the spacing between these peaks obeys an interval rule and determine the quantum number F associated with each peak.

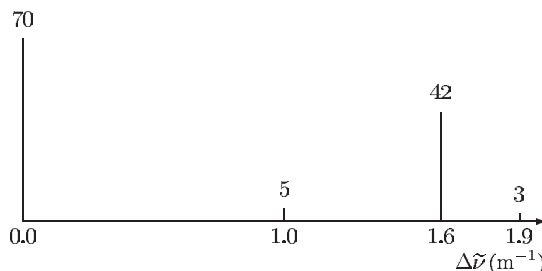
- (c) For the isotope ^{151}Eu , whose hyperfine structure was analysed in Example 6.2, the lower level of the transition has a hyperfine structure constant of $A(^8S_{7/2}) = 20 \text{ MHz}$ (measured by the method of magnetic resonance in an atomic beam (Sandars and Woodgate 1960)). What is the hyperfine-structure constant of this $^8S_{7/2}$ level for the isotope ^{153}Eu , analysed in this exercise?

(6.6) Interval for hyperfine structure

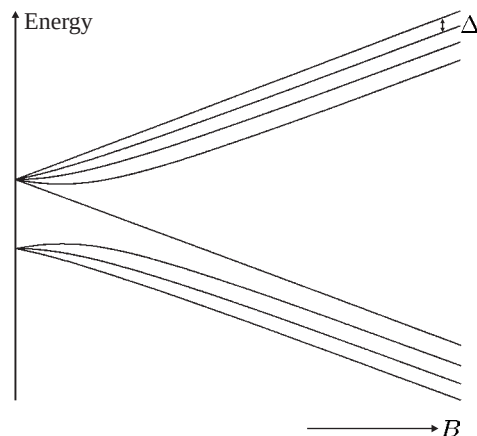
The $3d^5 4s 4p \ ^6P_{7/2}$ level of ^{55}Mn is split by hyperfine interaction into six levels that have separations 2599, 2146, 1696, 1258 and 838 MHz. Deduce the nuclear spin of ^{55}Mn and show that the separations confirm your value.

(6.7) Hyperfine structure

When studied by means of high-resolution spectroscopy, the resonance line $4s \ ^2S_{1/2} - 4p \ ^2P_{1/2}$ of naturally-occurring potassium consists of four components with spacings and intensity ratios as shown in the following diagram.



Natural potassium is a mixture of ^{39}K and ^{41}K in the ratio 14 : 1. Explain the origin of the structure, and deduce the nuclear spins and the ratio of the magnetic moments of the two isotopes.



(6.8) Zeeman effect on HFS at all field strengths

The figure shows the hyperfine structure of the ground level ($5s \ ^2S_{1/2}$) of ^{87}Rb (which has $A/h = 3.4 \text{ GHz}$), as a function of the magnetic flux density B .

- (a) Deduce the nuclear spin of this isotope of rubidium.
- (b) What are the appropriate quantum numbers for the states in both strong and weak fields (mark these on a copy of the figure)?
- (c) Show that in the weak-field regime the separation between states is the same in the upper and lower hyperfine levels.
- (d) In a strong field the energy of the states is given by eqn 6.33. Show that in this regime the four uppermost states have the same separation between them (marked Δ on the figure) as the four lower-lying states.

- (e) Define what is meant by a ‘strong field’ when considering hyperfine structure. Give an approximate numerical value for the magnetic field at which the cross-over from the weak-field to the strong-field regime occurs in this example.

(6.9) *Isotope shift*

Estimate the contributions to the isotope shift between $^{85}_{37}\text{Rb}$ and $^{87}_{37}\text{Rb}$ that arise from the mass and volume effects for the following transitions:

- (a) 5s–5p at a wavelength of ~ 790 nm; and
 (b) 5p–7s at a wavelength of ~ 730 nm.

Estimate the total isotope shift for both transitions, being careful about the sign of each contribution.

(6.10) *Volume shift*

Calculate the contribution of the finite nuclear size effect to the Lamb shift between the 2p $^2\text{P}_{1/2}$ and 2s $^2\text{S}_{1/2}$ levels in atomic hydrogen (using the information in Section 6.2.2). The measured value of the proton charge radius has an uncertainty of 1% and the Lamb shift is about 1057.8 MHz. What is the highest precision with which experimental measurement of the Lamb shift can test quantum electrodynamics (expressed as parts per million)?

(6.11) *Isotope shift*

Estimate the relative atomic mass A for which the volume and mass effect give a similar contribution to the isotope shift for $n^* \sim 2$ and a visible transition.

(6.12) *Specific mass shift*

An atom with a nucleus of mass M_N and N elec-

trons has a kinetic energy T given by

$$T = \frac{\mathbf{p}_N^2}{2M_N} + \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_e},$$

where $\mathbf{p}_N$ is the momentum of the nucleus and $\mathbf{p}_i$ is the momentum of the i th electron. The total of these momenta is zero in the centre-of-mass frame of the atom:

$$\mathbf{p}_N + \sum_{i=1}^N \mathbf{p}_i = 0.$$

Use this equation to express T in terms of electronic momenta only.

Answer the following for either (a) a lithium atom (with $N = 3$) or (b) the general case of a multi-electron atom with a nucleus of finite mass (i.e. any real non-hydrogenic atom). Find the kinetic-energy terms that are $\sim m_e/M_N$ times the main contribution: a normal mass effect (cf. eqn 6.21) and a specific mass effect that depends on products of the momenta $\mathbf{p}_i \cdot \mathbf{p}_j$.

(6.13) *Muonic atom*

A muon of mass $m_\mu = 207m_e$ is captured by an atom of sodium ($Z = 11$). Calculate the radius of the muon’s orbit for $n = 1$ using Bohr theory and explain why the atomic electrons have little influence on the energy levels of the muonic atom. Calculate the binding energy of the muon for $n = 1$. Determine the volume effect on the 1s–2p transition in this system; express the difference between the frequency of the transition for a nucleus with a radius r_N (given by eqn 6.25) and the theoretical frequency for $r_N = 0$ as a fraction of the transition frequency.

Web site:

<http://www.physics.ox.ac.uk/users/foot>

This site has answers to some of the exercises, corrections and other supplementary information.